Research Reproducibility Summary

Companion to the VaR/ES paper | 25 October 2026

This package accompanies Dynamic Joint Calibration of Combined VaR and ES Forecasts: Evidence from US Industries and the Yen. It records the data, code and numerical checks for the 26-page English manuscript. The checks were run on an earlier build of the same paper (its SHA-256 is in the README); the final manuscript differs only in its title page, one added funding sentence and layout, with identical tables and numbers.

What was reproduced

The models in both studies were re-estimated from the original inputs using the fixed settings, including the 36-series Study 1 panel, the historical-yen models, calibration, backtests and 9,999-draw bootstrap inference. Study 1 completed in 1571.1 seconds; its ten numerical CSV outputs were compared with the archive. No parameters were changed. All checked values agree at the precision reported in the manuscript.

Paper result	Reference value	Replay assessment
Industries: EW+DJC minus EW	-0.00787722842361139; Holm p = .0861	Exact agreement
Industries: H2 / H3	Holm p = .0006 / .0478	Exact agreement
Industries: EW+DJC violation rate	Approximately 2.1778%	Below the 2.25% tolerance bound
Historical yen: EW+DJC minus EW	-0.007068039844322028; p = .1973	Exact agreement

EW+DJC does not significantly improve the FZ0 score over EW in either sample. In the industry sample, H2 and H3 are supported under the prespecified block lengths, although H3 is sensitive to block length. Neither result establishes an improvement over EW. The risk calibration criterion is not met.

Numerical and presentation checks

The delivered replay comparison covers 427 CSV columns: 415 are exactly equal and 427 satisfy an absolute tolerance of 1e-12 (zero relative tolerance). All 106 saved historical-yen arrays are exactly equal to their archived counterparts. All 580 final-paper table cells match the supplied PDF after display normalization. The audit covers 13 tables, 105 body rows and 49 additional numerical claims; the five final-paper figures were regenerated from their retained inputs.

Data and software

Study 1 uses French 30-industry daily value-weighted returns and six Yahoo Finance index close series. Study 2 uses FRED DEXJPUS for 1974-2009. All eight evaluation input hashes match their original receipts. CPython 3.11.9 and exact package versions are recorded; the numerical core uses NumPy 2.4.3, pandas 3.0.1, SciPy 1.17.1 and arch 8.0.0.

FRED historical-yen data are included with attribution. French and Yahoo raw inputs are omitted because redistribution permission was not established. Exact access URLs, file structure, expected hashes and an input-verification utility are provided. The saved results can be checked and the tables and figures regenerated without these external inputs.

Verification Scope and Remaining Limits

Information timing

The inspected estimation, model-selection and calibration paths exclude current/future returns when producing each forecast. All 38 finite future-perturbation checks passed with exact equality for earlier forecasts, covering base forecasts, MCS, batch calibration, DJC experts/weights and Ye-type updates. These checks are evidence for the tested paths, not a formal proof for every possible input.

The timing checks condition on the frozen design. Historical-yen validation runs backward in calendar time relative to method development, and Study 1 development/evaluation histories overlap across series. Neither design establishes prospective historical deployment.

Appreciation-state classification and full-sample tertile thresholds are explicitly retrospective diagnostics. They must not be read as ex-ante signals. Later-vintage historical data also do not establish point-in-time data availability.

Execution outside the source repository

A new Python 3.11.9 virtual environment installed the 17 packages at the recorded versions and passed pip check. The package was copied outside the source repository and run with isolated Python imports. Study 2 was refitted and its inference, tables and figures were regenerated successfully. The full Study 1 pipeline also completed there in 1419.8 seconds after authorized inputs were imported and hash-verified. All commands in the isolated replay sequences returned exit status 0. The package therefore executed without imports from the original research checkout.

Items that cannot be directly verified

1. Original Study 1 per-series daily VaR/ES paths were not archived. Fresh estimation can be compared with saved daily panel scores and per-series backtest statistics, but direct equality to those missing original paths cannot be established.

2. A fully self-contained distributed Study 1 refit is unavailable without lawful access to the exact external raw inputs. Fresh provider downloads may differ. Local successful replay and unrestricted public-data availability are separate claims.

3. Point-in-time revisions and constituent survivorship properties are outside the retained evidence, and the order of events before each protocol freeze rests on commit records kept by the author. The current development score files reproduce the frozen block lengths but lack original freeze-date hash receipts. Development was not retuned or rerun as a new confirmation.

4. Cross-platform optimizer equivalence has not been tested. Numerical agreement does not by itself establish statistical validity.

Suggested reading and execution order

Read README.md and Data_Sources.pdf, then outputs/verification/Verification_Report.md. Inspect numeric_comparison.csv and paper_cell_comparison.csv for the detailed comparisons; provenance/code_changes.json records all code edits and file hashes; provenance/Prespecified_Design.md summarizes the frozen protocols. The README provides commands to create a fresh workspace, refit Study 2, supply authorized Study 1 inputs, regenerate tables/figures and compare all outputs.

The ZIP has a file-level SHA-256 manifest and an external archive receipt. Original source-project files and the completed manuscript were preserved.